

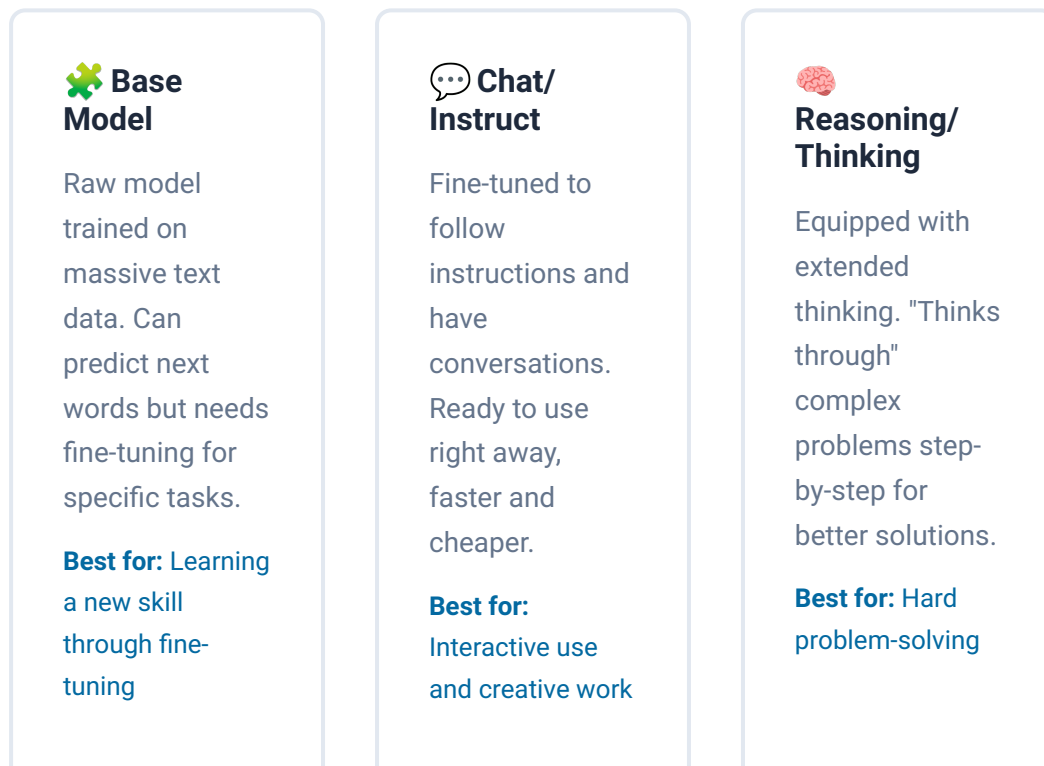
# Understanding LLMs

From Base Models to Frontier AI

*Your journey into Large Language Models starts here*

# LLMs Come in 3 Flavors

Think of these as three different versions of the same core technology, each optimized for different jobs:



## WHY THREE TYPES?

Different tasks have different needs. You don't use a truck for precision work, and you don't use a sports car to move houses. Same with LLMs—choose the right tool for the job.

# Base Models: The Raw Foundation

A Base Model is an LLM trained purely on predicting the next word in a sequence. It's the raw engine.

## HOW IT WORKS

Trained on billions of words from books, articles, code, etc. The model learns: "When I see X, Y usually follows." It's like an advanced autocomplete—type a few words, it guesses the next ones.

## Strengths

- Can be fine-tuned for ANY specific task
- Most flexible foundation
- Good for learning domain-specific language
- Lower inference cost baseline

## Limitations

- Can't follow instructions well out-of-the-box
- Might generate random or weird text
- Requires fine-tuning to be useful
- Not ready for production use

### **Real Example**

If you're building a medical AI assistant, you'd start with a Base Model, then fine-tune it on medical literature so it learns medical terminology and context.

# Chat & Instruct Models: Ready to Talk

A Chat Model is a Base Model that's been fine-tuned to understand instructions and have conversations. It's "trained to be helpful."



## HOW IT'S DIFFERENT

After training on raw text, engineers give the model feedback on which responses are good and which are bad. Through this process (called RLHF - Reinforcement Learning from Human Feedback), it learns to follow instructions and chat naturally.



### Why We Use Them

- Works immediately out-of-the-box
- Faster and cheaper than reasoning models
- Perfect for chatbots and content generation
- Safe—already aligned to be helpful



### Trade-offs

- Less flexible than Base Models
- Might hallucinate on complex topics
- Not optimized for deep reasoning
- Fixed personality from training

## Real Examples

**ChatGPT, Claude, Gemini** are all Chat Models. When you ask them to "write a poem" or "explain quantum physics," that's chat model territory. Fast, practical, creative.

# Reasoning Models: The Deep Thinkers

A Reasoning Model is designed to "think through" hard problems step-by-step, spending more compute upfront to get better answers.

## "BUDGET FORCING" EXPLAINED

The model is trained with "budget forcing"—by artificially adding thinking time, it performs better at complex tasks. Like giving a student extra time on a hard exam, they can think deeper and get better answers.

### What You Get

- Solves hard math/logic problems
- Better coding for complex tasks
- Can reason through multi-step problems
- Higher accuracy on tough questions

### The Cost

- Slower (needs time to think)
- More expensive (more compute)
- Overkill for simple tasks
- Visible thinking (longer responses)

### **Real Examples**

When debugging complex code, solving a challenging algorithm problem, or working through a math proof—you'd want a Reasoning Model. It's worth the extra time and cost for hard problems.

# Frontier Models: Who's Leading?

These are the most advanced LLMs available today, each from major AI companies:

## Anthropic

### Claude Models

Known for safety, reasoning, and detailed analysis. Strong at writing and code.

## OpenAI

### GPT Models

Most popular. Great at creative writing, coding, and wide range of tasks.

## Google

### Gemini Models

Multimodal (text + images). Integrates well with Google services.

## THE COMPETITION

There's also x.ai (Grok), DeepSeek, and others. The field is competitive. What matters: For your specific task, which model performs best and fits your budget?

 **NO CLEAR WINNER**

Each excels at different things. Choose based on: accuracy for your task, cost, speed, and specific capabilities you need. It's like picking between different tools in a toolbox.

# Why Frontier LLMs Are Mind-Blowing

What can these models actually do well?



## Synthesizing Information

Take complex topics and explain them clearly. Answer questions in depth with structured, researched answers including summaries.



## Fleshing Out Ideas

Give rough notes, it builds a well-crafted email or blog post. Iterates with you until it's perfect.



## Coding

Write and debug code. Far overtaken Stack Overflow as the go-to resource for engineers solving problems.



## Real Student Example

You're stuck on a coding assignment. Instead of searching Stack Overflow for 30 minutes, you ask Claude/ChatGPT to explain the error and suggest fixes. You get a response in seconds, understand it better, and learn the concept.

### THE IMPACT

These models have genuinely changed how people work. They're not perfect, but they're incredibly useful assistants for research, writing, coding, and problem-solving.

# Limitations: What They CAN'T Do

Frontier models are powerful, but they have real constraints:



## Specialized Domains

Not PhD-level in niche fields. Limited knowledge in highly specialized domains they weren't trained on.



## Recent Events

Knowledge cuts off at training date. Can't access current news or real-time information without external tools.



## Mistakes & Hallucinations

Can confidently give wrong answers. May make blind jumps to conclusions instead of reasoning carefully.



## REAL LIMITATION

**Hallucinations are tricky:** When a model doesn't know something, it might make up an answer that sounds plausible. This is why you should always verify important information.



### KEY LESSON

These models are tools, not oracles. They're best used as assistants to boost your productivity, not as the final source of truth. Always verify. Always think critically.

# Choosing the Right Model for Your Task

Here's a practical decision framework:

## ? ASK YOURSELF THESE QUESTIONS

1. **How complex is the task?** (Simple = Chat, Complex = Reasoning)
2. **What's your budget?** (Reasoning is more expensive)
3. **How much time do I have?** (Reasoning is slower)
4. **Do I need latest info?** (Might need external data integration)
5. **What accuracy level do I need?** (Higher stakes = need better verification)

### ✓ Choose Chat Model When:

- Brainstorming ideas
- Writing content
- Explaining concepts
- Quick coding help
- Time is limited

### ✓ Choose Reasoning Model When:

- Solving complex algorithms
- Deep debugging needed
- Multi-step problem solving
- High accuracy required
- Time/cost allows

# Real-World Use Cases for Students

---

How you can actually use this stuff right now:



## Assignment Help

Use Chat Models to brainstorm essay outlines, explain difficult concepts, or debug your code. Then use your brain to verify and refine.

[ChatGPT](#)

[Claude](#)



## Learning to Code

Stuck on a bug? Paste the error message and your code. The model explains what went wrong and how to fix it. Learn by understanding the explanation.

[Coding Assistants](#)



## Building AI Projects

Want to build something like a chatbot or RAG system? These models are your building blocks. Use Claude API or OpenAI API to integrate them into your projects.

[API Access](#)

 **ACADEMIC INTEGRITY**

Using AI to learn is great. Using it to cheat is not. Check your school's policy. Best practice: Use AI to understand and learn, not to submit as your own work.

# Advanced Features: Deep Research & Agents

---

Some models like Claude have powerful advanced capabilities:



## Deep Research Mode

Claude can plan a research task, execute multiple parallel searches, fetch and analyze sources, cross-reference information, and provide a synthesized, cited report. It's like having a research assistant.



## Agent Mode

Claude can take complex tasks (like "book a restaurant for 4 at 9PM in Bangalore with masala dose menu"), break them down, use tools, and execute multi-step workflows to get things done.



## For Your RAG Project

Deep Research could gather and synthesize information for your project automatically. Agents could handle complex multi-step workflows. These are the "next level" of LLM capabilities.

### THE FUTURE

As models get better, they'll do more than chat. They'll research, plan, execute, and build. The students who understand these tools will have a significant advantage in any field.

Slide 12 of 12

# Your AI Journey Starts Now

Understanding these 3 flavors and frontier models is just the beginning.

- ✓ Base Models: Raw foundation for fine-tuning
- ✓ Chat Models: Ready-to-use, fast, practical
- ✓ Reasoning Models: Deep thinking for hard problems
- ✓ Frontier LLMs: Frontier models are remarkable
- ✓ Use tools wisely: Always verify, stay critical
- ✓ Build with them: APIs let you create amazing things

Slide 1

← Previous

Next →

*The future is collaborative: AI + human intelligence*

Slide 1

← Previous

Next →